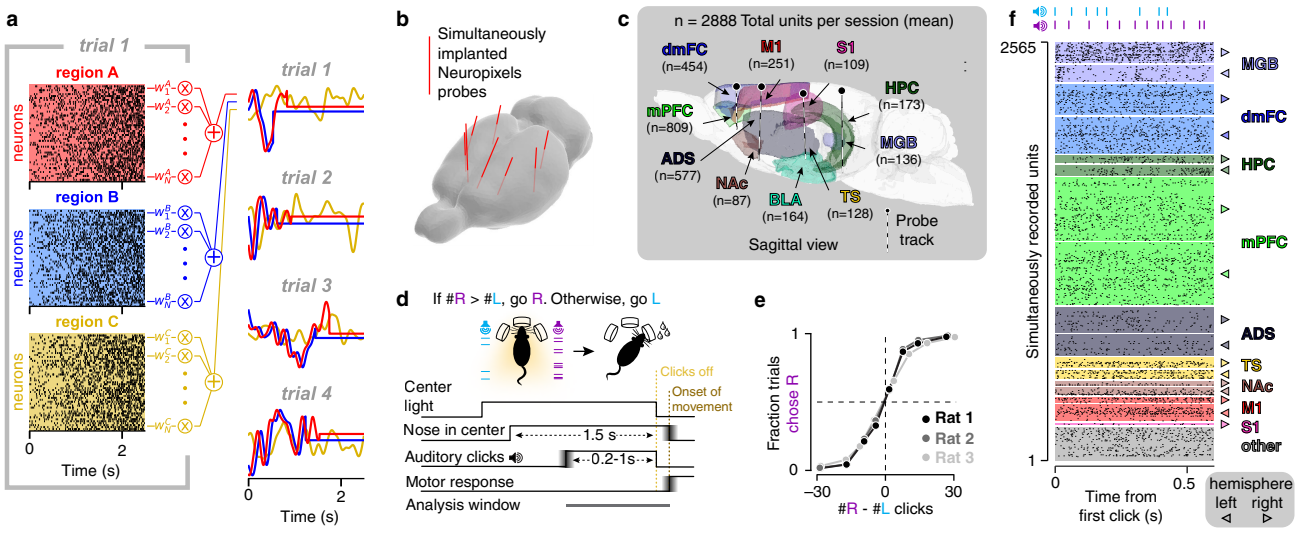

a
trial 1
region A
region B
region C
neurons
neurons
neurons
Time (s)
trial 1
trial 2
trial 3
trial 4
Time (s)
b
Simultaneously implanted Neuropixels probes
c
n = 2888 Total units per session (mean)
dmFC (n=454)
M1 (n=251)
S1 (n=109)
HPC (n=173)
mPFC (n=809)
MGB (n=136)
ADS (n=577)
NAc (n=87)
BLA (n=164)
TS (n=128)
Sagittal view
Probe track
d
If #R > #L, go R. Otherwise, go L
Clicks off
Onset of movement
Center light
Nose in center
1.5 s
Auditory clicks
0.2-1s
Motor response
Analysis window
e
Fraction trials chose R
Rat 1
Rat 2
Rat 3
#R - #L clicks
f
2565
1
Simultaneously recorded units
MGB
dmFC
HPC
mPFC
ADS
TS
NAc
M1
S1
other
Time from first click (s)
hemisphere
left
right